ASSOCIATIVITY OF THE RELATIVE CUP PRODUCT FOR A MORPHISM OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the commutativity of the diagram: the transposition is displayed and the two sides are identified in the same display.

Editorial note. The single labelled reference, which occurs inside the counted statement, is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space, and where applicable in chapter more-algebra for a ring. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here; the special-case twins are to be left uncounted by the lanes holding those chapters.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros. The Xy-pic diagram inside the counted statement is delivered as the author wrote it.

Allowed standard prerequisites (editorial).

- Relative cup product for a morphism of ringed topoi. Same chapter, the morphism Rf_*K tensor^L Rf_*L -> Rf_*(K tensor^L L) named in the counted statement. Supplies the two arrows whose composite is asserted to be independent of the order of bracketing.
- Derived pullback Lf^* and the projection-style adjunction with Rf_*. Same chapter, the adjunction between Lf^* and Rf_* on derived categories of modules. Supplies the passage to adjoints with which the proof replaces the comparison of two composites by the comparison of two maps in D(O).
- Monoidality of the derived pullback. Same chapter, the identification Lf^*(A tensor^L B) = Lf^*A tensor^L Lf^*B used in the displayed computation. Supplies the first equality in the displayed chain, which produces the common adjoint map.

Source reference. Original remark 0B6C: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4193–4193. Same-chapter navigation pointer occurring inside the counted statement, where the author names the relative cup product of that remark. The pointer only resolves the reference for the reader; the remark is not reproduced inside the counted span, and no bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. The relative cup product of Remark 0B6C is associative in the sense that the diagram*

$$\begin{array}{ccc} Rf_*K \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_*L \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_*M & \longrightarrow & Rf_*(K \otimes^{\mathbf{L}}_{\mathcal{O}} L) \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_*M \\ \downarrow & & \downarrow \\ Rf_*K \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_*(L \otimes^{\mathbf{L}}_{\mathcal{O}} M) & \longrightarrow & Rf_*(K \otimes^{\mathbf{L}}_{\mathcal{O}} L \otimes^{\mathbf{L}}_{\mathcal{O}} M) \end{array}$$

is commutative in $D(\mathcal{O}')$ for all K, L, M in $D(\mathcal{O})$.

Proof. Going around either side we obtain the map adjoint to the obvious map

$$\begin{aligned} Lf^*(Rf_*K \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_*L \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_*M) &= Lf^*(Rf_*K) \otimes^{\mathbf{L}}_{\mathcal{O}} Lf^*(Rf_*L) \otimes^{\mathbf{L}}_{\mathcal{O}} Lf^*(Rf_*M) \\ &\to K \otimes^{\mathbf{L}}_{\mathcal{O}} L \otimes^{\mathbf{L}}_{\mathcal{O}} M \end{aligned}$$

in $D(\mathcal{O})$. □